

MP3 Tag Reader

Introduction

A console-based application in C which reads and displays metadata (ID3 tags) from MP3 files and also allows editing of tag fields like title, artist, album, year, track, genre and comment. It reads the tag data directly from the file bytes.

Data used

1. **ID3v2.3 tag structure** – The tag is made up of frames, where each frame has an ID (like TIT2 for title), a size and the actual text. The program reads each frame one by one using its size to know where it ends and where the next one starts.
2. **Temporary file for editing** – Since a new value can be a different length than the old one, editing is done by writing a new file with the updated frame, instead of changing the file in place.

Files

1. **main.c** – checks the command line arguments and calls view or edit based on what is asked
2. **check_version.c** – checks if the file has an ID3 tag
3. **viewmp3.c** – reads and prints all the tag details
4. **editmp3.c** – edits a tag field and writes the updated file
5. **mp3_tag_reader.h** – common function declarations used by all files

Features

1. view all tag details – title, artist, album, year, track, genre, comment
2. shows which ID3 version the file has
3. edit title, artist, album, year, track, genre and comment
4. checks if the command line arguments and file are valid
5. shows a help menu
6. shows an error if no tag is found in the file

Sample output

```
shreya@LAPTOP-2Q70AT88:~/cprograms/mp3_tag_reader_project$ ./mp3_tag_reader.out -v sample.mp3
ID3 v2.3:
-----MP3 TAG DETAILS-----
Title: Sunny Sunny - Yo Yo Honey Singh - [SongsPk.CC]
Artist: Yo Yo Honey Singh - [SongsPk.CC]
Album: Yaariyan
Year: 2013
Genre: Bollywood Music - [SongsPk.CC]
Comment: Downloaded From SongsPk.CC
-----
```

```
shreya@LAPTOP-2Q70AT88:~/cprograms/mp3_tag_reader_project$ ./mp3_tag_reader.out -e -y 2026 sample.mp3
Editing TYER...
Old value : 2013
New value : 2026

After editing:
ID3 v2.3:
-----MP3 TAG DETAILS-----
Title: Sunny Sunny - Yo Yo Honey Singh - [SongsPk.CC]
Artist: Yo Yo Honey Singh - [SongsPk.CC]
Album: Yaariyan
Year: 2026
Genre: Bollywood Music - [SongsPk.CC]
Comment: Downloaded From SongsPk.CC
-----
```

Conclusion

This project reads and edits ID3 tags in an MP3 file by directly working with the file bytes. Building it helped in understanding how a real file format is structured and how to edit part of a file without corrupting the rest of it.